

~~CONFIDENTIAL~~
SECURITY CLASSIFICATION

NSA STAFF PROCESSING FORM

TO DDO	EXREG CONTROL NUMBER	KCC CONTROL NUMBER	Z CONTROL NUMBER Z-054-98
THRU		ACTION ☐ APPROVAL ☐ ACTION ☒ INFORMATION	EXREG SUSPENSE KCC SUSPENSE ELEMENT SUSPENSE
SUBJECT CIA KRYPTOS Sculpture -- The Challenge and Resolution (U)			
DISTRIBUTION Z4 , Z23			

SUMMARY

(b) (3)-P.L. 86-36

1. (U) In response to your request we have put together a package chronologically outlining the events in our decryption (89%) of the CIA courtyard KRYPTOS sculpture. Also attached is a copy of the original memorandum to Adm. McConnell with attachments providing the cipher, the cryptography employed, and the respective decrypts.

2. (U) The initial examination of the cipher revealed that it was likely to consist of three cryptographically distinct sections. Basic computer diagnostic tools confirmed this hypothesis. Subsequent analysis and solution, however, did not require any compute power.

3. (C) Parts 1-3 were solved within two days of receiving the informal tasking from Chief, Z. Another day was spent on the final section and a decision was made to stop any further work. Given the suspected cryptography, the last section is too short to solve without diverting a great deal of effort from operational problems.

(b) (3)-P.L. 86-36

COORDINATION/APPROVAL							
OFFICE	NAME AND DATE		SECURE PHONE	OFFICE	NAME AND DATE		SECURE PHONE
Z09		7/22	5043s				
				Drafter			4451s
				ORG.	PHONE (Secure)	DATE PREPARED	

Chronological history of NSA personnel and their involvement
in the partial decryption of the KRYPTOS sculpture
located in the CIA courtyard

- 1988 - The CIA Fine Arts Commission approves James Sanborn's proposal.
- 1990 - The artwork, titled *Kryptos*, is dedicated. A portion of this work of art consists of a classic Vigenere Square and 870 characters of cipher punched through two large copper sheets.
- 1991 - While on a trip to the CIA headquarters, an informal group comprised mainly of Cryptanalysis interns, handwrites the cipher onto sheets of paper, and distributes it to any and all interested cryptanalysts back at NSA.
- 1992 - Official challenge for solution is relayed through DCI at a Gold Bug award ceremony.
- 1992 - Mr. [] is the first person to decrypt a portion of the cipher. The cipher system used is a polyalphabetic substitution using eight alphabets. The decrypted text accounts for the last 373 characters from the first section of 436, but the initial 63 characters resist decryption. Because of this, analysts concede that four distinct sections are likely, with this being the second section. (b) (3)-P.L. 86-36
- 1992 - Mr. [] is the second person to successfully decrypt a portion of the cipher. That portion is the third section involving 337 characters and employing a transposition system using a matrix with dimensions 4 X 86.
- 1992 - As the third successful cryptanalyst, Mr. [] successfully decrypts the initial 63 characters of the cipher, which is the first section. It also uses a polyalphabetic substitution, but with 10 alphabets.
- 1992 - An informal document is produced detailing the solution of the three sections. These three sections comprise the first 773 characters out of 870 total, leaving the last 97 characters unresolved.
- 1993 - A formal letter is sent to Adm. McConnell (DIRNSA) detailing the story, and is returned with a request that it be forwarded to Adm. Studeman at CIA.
- 1998 - There is renewed interest from the CIA, with an eye towards a technical article for an internal publication.

DONS:
Zy23

142-93

UNITED STATES GOVERNMENT

memorandum

Z-043-93

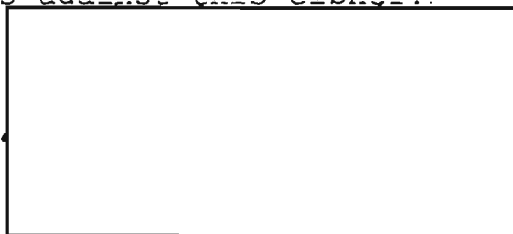
DATE: 9 June 1993

REPLY TO
ATTN OF: ZPLS FORWARD TO ADMIRAL
STUDEMAN FOR INFO.SUBJECT: CIA KRYPTOS Sculpture - Challenge and Resolution (FOUO)
INFORMATION MEMORANDUMTO: DIR M
THRU: D/DIR JH, EXEC/DIR DD, DDOGreat
Story!

1. ~~(FOUO)~~ The KRYPTOS sculpture, located at the entrance and in the courtyard of the new CIA headquarters, consists of a series of stone "pages" containing code which begins as International Morse and increases in complexity as the stonework extends into the courtyard. Inserted between these stone "pages" is a flat copper sheet engraved with letters and symbols - the enciphered message - that is the focus of this challenge.

2. ~~(FOUO)~~ In November, a cadre of cryptanalysts assigned to Z Group enthusiastically responded to the challenge. Within one month, three of the four cipher systems used to encrypt the sculpture's plain text had been diagnosed and completely exploited. The cryptographies employed for the encryption of these three parts involved two periodic polyalphabetic substitution ciphers and a keyed columnar transposition cipher. The exploitation of the sculpture's first three parts constitutes a readability of approximately 99%. The final 97 characters continue to elude solution.

3. ~~(FOUO)~~ Attached, for your review, is a brief description of the employed cryptographies and the plain text derived from the three exploited portions of the KRYPTOS sculpture. If your schedule permits, we would be happy to present a 15-minute briefing on the KRYPTOS sculpture solution and introduce you to the cryptanalysts responsible for the success against this cipher.



3 Encls:

1. Copy of Sculpture Picture
2. Copy of Cipher
3. Description of Cryptographies

(b) (3)-P.L. 86-36

cc: Z4
Z43

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OPTIONAL FORM

46695



6-130

THE KRYPTOS SCULPTURE CIPHER

E M U F P H Z L R F A X Y U S D J K Z L D K R N S H G N F I V J
 Y Q T Q U X Q B Q V Y U V L L T R E V J Y Q T M K Y R D M F D
 V F P J U D E E H Z W E T Z Y V G W E K K Q E T G F Q J N C E
 G G W H K K 2 D Q M C P F Q Z D Q M M I A G P F X H Q R L G
 T I M V M Z J A N Q L V K Q E D A G D V F R P J U N G E U N A
 Q Z G Z L E C G Y U X U E E N J T B J L B Q C R T B J D F H J R R
 Y I Z E T K Z E M V D U F K S J H K F W H K U W Q L S Z F T I
 H H D D D U V H 2 D W K B F U F P W N T D F I Y C U Q Z E R E
 E V L D K F E Z M O Q Q J L T T U G S Y Q P F E U N L A V I D X
 F L G G T E Z 2 F K Z B S F D Q V G O G I P U F X H H D R K F
 F H Q N T G P U A E C N U V P D J M Q C L Q U M U N E D F Q
 E L Z Z V R R G K F F V O E E X B D M V P N F Q X E Z L G R E
 D N Q F M P N Z G L F L P M R J Q Y A L M G N U V P D X V K P
 D Q U M E B E D M H D A F M J G Z N U P L G E W J L L A E T G
 E N D Y A H R O H N L S R H E O C P F E O I B I D Y S H N A I A
 C H T N R E Y U L D S L L S L L N O E S N O S M R W X M N E
 T P R N G A T I H N R A R P E S L N N E L E B L P I I A C A E
 W M T W N D I T E E N R A H C T E N E U D R E T N H A E O E
 T F O L S E D T I W E N H A E I O Y F E Y Q H E E N C T A Y C R
 E I F T B R S P A M H H E W E N A T A M A T E G Y E E R L B
 T E E F O A S F I O T U E T U A E O F O A R M A E E R T N R T I
 B S E D D N I A A H T T M S T E W P I E R O A G R I E W F E B
 A E C T D D H I L C E I H S I T E G O E A O S D D R Y D L O R I T
 R K L M L E H A G T D H A R D P N E O H M G F M F E U H E
 E C D M R I P F E I M E H N L S S T E R T V D O H W 2 O B K R
 U O X O G H U L B S O L I F B B W F L R V Q Q P R N G K S S O
 T W T Q S J Q S S E K Z Z W A T J K L U D I A W I N F B N Y P
 V T T M Z F P K W G D K Z X T J C D I G K U H U A U E K C A R

PART 1

Cryptography: Periodic Polyalphabetic Substitution employing
10 alphabets

Plain component: Keyword mixed sequence based on KRYPTOS

Cipher component: Keyword mixed sequence based on KRYPTOS

Repeating Key: PALIMPSEST

Index letter: K

P:	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z
C1:	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
C2:	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S
C3:	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J
C4:	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H
C5:	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L
C6:	P	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y
C7:	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
C8:	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O	S	A	B	C	D
C9:	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P	T	O
C10:	T	O	S	A	B	C	D	E	F	G	H	I	J	L	M	N	Q	U	V	W	X	Z	K	R	Y	P

EMUFPHZLRF AXYUSDJKZL DKRNSHGNFI VJYQTQUXQB
BETWEENSUB TLESHADING ANDTHABSCE NCEOFLIGHT

QVYUVLLTRE VJYQTMKYRD MFD
LIESTHENUA NCEOFIQLUS ION

(FOUO) Respaced and punctuated, it reads:

"BETWEEN SUBTLE SHADING AND THE ABSENCE OF LIGHT LIES THE NUANCE OF
ILLUSION"

PART 2

Cryptography: Periodic Polyalphabetic Substitution employing
8 alphabets

Plain component: Keyword mixed sequence based on KRYPTOS

Cipher component: Keyword mixed sequence based on KRYPTOS

Repeating Key: ABSCISSA

Index letter: K

P: KRYPTOSABCDEFGHIJLMNQUVWXZ
C1: ABCDEFGHIJLMNQUVWXZKRYPTOS
C2: BCDEFGHIJLMNQUVWXZKRYPTOSA
C3: SABCEDEFGHIJLMNQUVWXZKRYPTO
C4: QDEFGHIJLMNQUVWXZKRYPTOSAB
C5: IJLMNQUVWXZKRYPTOSABCDEFGHI
C6: SABCEDEFGHIJLMNQUVWXZKRYPTO
C7: SABCEDEFGHIJLMNQUVWXZKRYPTO
C8: ABCDEFGHIJLMNQUVWXZKRYPTOS

VFPJUDEE HZWETZYV GWHKKQET GFQUNCEG GWHKK?DQM
ITWASTOT ALLYINVI SIBLEHOW STHATPOS SIBLE?THE

CPFQZDQM MIAGPEXH QRLGTIMV MZJANQLV KQEDAGDV
YUSEDTHE EARTHSMA GNETICFI ELDXTHEI NFORMATI

FRPJUNGE UNAQZGZL ECGYUXUE ENJTBULB QCETBJDF
ONWASGAT HEREDAND TRANSMIT TEDUNDER GROUNDTO

HRRYZET KZEMVDUF KSJHKFWH KUWQLSZF TIHHDDDU
ANUNKNOW NLOCATIO NXDOESLA NGLEYKNO WABOUTTH

VH?DWKBFU FPWNTDKI YCUQZERE EVLDKFEZ MOQQJLTT
IS?THEYSH OULDITSB URIEDOUT THERESOM EWHEREXW

UGSYQPFE UNLAVIDX FLGGTEZ?F KZESFDQV GOGIPUFX
HOKNOWST HEEEXACTL OCATION?O NLYWWTHI SWASHISL

HADRKEFH QNTGPAUE CNUVPDJM QCLQUMUN EDFQELZZ
ASTMESSA GEXTHIRT YEIGHTDE GREESFIF TYSEVENM

VRRGKFFV OEEXBDMV PNFOXZEL GREDNQFM PNZGLFLP
INUTESSI XPOINTFI VESECOND SNORTHSE VENTYSEV

PART 3

Cryptography: Keyed Columnar transposition

Matrix size: Incompletely filled 4 X 86

Specific key: KRYPTOS, numerically keyed and repeated 13 times
(first 12 columns listed below)

Route: Bottom to top

SLOWLYDESPARATLYSLOWLYTHEREMAINSOFPASSAGEDE
ASREMOVEDWITHTREMBLINGHANDSIMADEATINYBREACH
OLEALITTLEIINSERTEDTHECANDLEANDPEEREDINTHEH
FLICKERBUTPRESENTLYDETAILSOFTHEROOMWITHINEM

1	1	1	9	8	7
2	1	0			

BRISTHATENCUMBEREDTHELOWERPARTOFTHEDOORWAYW
INTHEUPPERLEFTHANDCORNERANDTHENWIDENINGTHEH
OTAIRESCAPINGFROMTHECHAMBERCAUSEDTHEFLAMETO
ERGEDFROMTHEMISTXCANYOUSEEANYTHINGO

6	5	4	3	2	1
---	---	---	---	---	---

"SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED. WITH TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER, AND THEN, WIDENING THE HOLE A LITTLE, I INSERTED THE CANDLE AND PEERED IN. THE HOT AIR ESCAPING FROM THE CHAMBER CAUSED THE FLAME TO FLICKER, BUT PRESENTLY, DETAILS OF THE ROOM WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q".

Note: The above is a paraphrase from The Tomb of Tut-Ankh-Amen written by Mr. Howard Carter.